南方科技大学
SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY

Course Name: **General Physics B (II)** **Department:** **PHYSICS**

Exam Duration: **2 hours** **Exam Paper Setter:** **Physics teaching team**

Question No.	1	2	3	4	5	6	7	8	9	10
Score										

This exam paper contains 8 questions and the score is 100 in total. (Hand in your exam paper, answer sheet, and draft paper to the proctor when the exam ends.)

Q1. Multiple Choice Questions ($3\times10=30$ scores), each question has one correct answer.

1.The electric field due to a uniform distribution of charge on a spherical shell is zero:

A) everywhere

B) nowhere

C) only at the center of the shell

D) only inside the shell

E) only outside the shell

2. The graph shows the electric potential as a function of x in a certain region. What is the x component of the electric field in this region if V_s = 50 V?

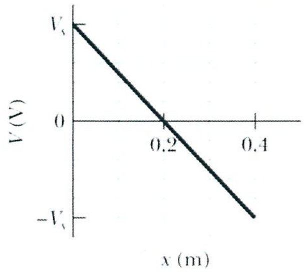
A) 250 V/m

B) 40 V/m

C) 10 V/m

D) –40 V/m

E) –250 V/m



3. Suppose the current charging a capacitor is kept constant. Which graph below correctly gives the potential difference V across the capacitor as a function of time?

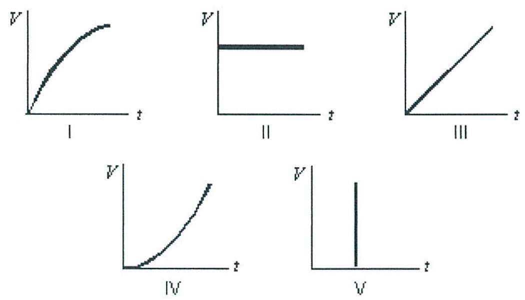
A) I

B) II

C) III

D) IV

E) V



4. Two long straight wires enter a room through a window. One carries a current of 3.0 A into the room while the other carries a current of 5.0 A out. The magnitude of the path integral $\oint \vec{B} \cdot d\vec{s}$ around the window frame is:

A) 2.5×10^{-6} T·m
B) 3.8×10^{-6} T·m
C) 6.3×10^{-6} T·m
D) 1.0×10^{-5} T·m
E) none of these

5. As a loop of wire with a resistance of 10 Ω moves in a non-uniform magnetic field, it loses kinetic energy at a uniform rate of 5 mJ/s. The induced emf in the loop is:
A) 0 V
B) 0.22 V
C) 0.28 V
D) 2.0 V
E) cannot be calculated from the given data

6. In a sinusoidally driven series *RLC* circuit, the inductive resistance is $X_L = 200\ \Omega$, the capacitive reactance is $X_C = 100\ \Omega$, and the resistance is $R = 50\ \Omega$. The current and applied emf would be in phase if:
A) the resistance is increased to 100 Ω, with no other changes
B) the resistance is increased to 200 Ω, with no other changes
C) the inductance is reduced to zero, with no other changes
D) the capacitance is doubled, with no other changes
E) the capacitance is halved, with no other changes

7. Light with an intensity of 1 kW/m^2 falls normally on a surface and is completely reflected. The radiation pressure is:
A) 1 kPa
B) 3×10^{11} Pa
C) 1.7×10^{-6} Pa
D) 3.3×10^{-6} Pa
E) 6.7×10^{-6} Pa

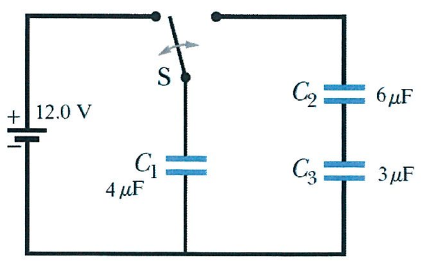
8. In a Young's double-slit experiment, a thin sheet of mica is placed over one of the two slits. As a result, the center of the fringe pattern (on the screen) shifts so that the center is now occupied by what was the 30th dark band. The wavelength of the light in this experiment is 480 nm and the index of the mica is 1.60. The mica thickness is:
A) 0.009 mm
B) 0.012 mm
C) 0.014 mm
D) 0.024 mm
E) 0.062 mm

9. At the first minimum adjacent to the central maximum of a single-slit diffraction pattern the phase difference between the Huygens wavelet from the top of the slit and the wavelet from the midpoint of the slit is:
A) $\pi/8$ rad
B) $\pi/4$ rad
C) $\pi/2$ rad
D) π rad
E) $3\pi/2$ rad

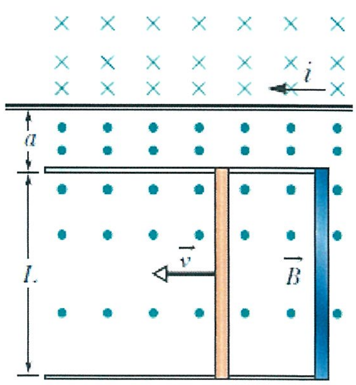
10. A meson when at rest decays 2 μs after it is created. If moving in the laboratory at $0.99c$, its lifetime according to laboratory clocks would be:
A) the same
B) 0.28 μs
C) 4.6 μs
D) 14 μs
E) none of these

Essay Questions
Q2. (6 scores) Find the maximum kinetic energy of electrons ejected from a certain material if the material's work function is 2.3 eV and the frequency of the incident radiation is 2.5×10^{15} Hz.

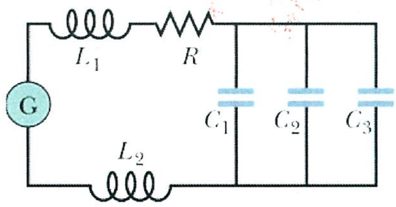
Q3. (10 scores) As shown in the figure, switch is thrown to the left side until C_1 is fully charged. Then the switch is thrown to the right. What is the final charge on C_1, C_2 and C_3?



Q4. (12 scores) The figure shows a rod of length $L = 10.0$ cm that is forced to move at constant speed v=5.00m/s along horizontal rails. The rod, rails, and connecting strip at the right form a conducting loop. The rod has resistance $0.400\,\Omega$; the rest of the loop has negligible resistance. A current $i = 100$ A through the long straight wire at distance a=10.0 mm from the loop sets up a (nonuniform) magnetic field through the loop. Find: (a) the emf and current induced in the loop. (b) At what rate is thermal energy generated in the rod? (c) What is the magnitude of the force that must be applied to the rod to make it move at constant speed? (d) At what rate does this force do work on the rod?



Q5. (12 scores) In the figure, a generator with an adjustable frequency of oscillation is connected to resistance R = 100 Ω, inductances L_1 = 9.70 mH and L_2 = 2.30 mH, and capacitances C_1 =8.40 mF, C_2 = 2.50 mF,and C_3 = 3.50 mF.(a) What is the resonant frequency of the circuit? What happens to the resonant frequency if (b) R is increased, (c) L_1 is increased, (d) C_3 is removed from the circuit,and (e) L_2 is removed?



Q6. (10 scores) In a double-slit arrangement the slits are separated by a distance equal to 100 times the wavelength of the light passing through the slits. (a) What is the angular separation in radians between the central maximum and an adjacent maximum? (b) What is the distance between these maxima on a screen 90.0 cm from the slits?

Q7. (12 scores) How many bright fringes appear between the first diffraction envelope minima to either side of the central maximum in a double-slit pattern if λ = 550 nm, d = 0.180 mm, and a = 30.0 mm? What is the ratio of the intensity of the third bright fringe to the intensity of the central fringe?

Q8. (8 scores) An armada (a large group of war ships) of spaceships that is 1.00 ly long (in its rest frame) moves with speed 0.850*c* relative to a ground station in frame *S*. A messenger travels from the rear of the armada to the front with a speed of 0.950*c* relative to *S*. How long does the trip take as measured

(a) in the armada's rest frame

(b) in the messenger's rest frame,